CHAPTER 6
DISCUSSION

This section interprets the measurements of Section 5 against the four challenges of Section 1.2 and the design of Section 3. It introduces no new measurement and no new metric: every figure below is taken from a table of Section 5 and carries the evidence class stated there (MEASURED, CALCULATED, SIMULATED, SOFTWARE-TEST, ESTIMATE or [PENDING MEASUREMENT]). Section 6.1 states what the evidence supports and the bound within which it holds, Section 6.2 sets out the design trade-offs and what the build and the audit taught, Section 6.3 lists the claims the paper does not make, and Section 6.4 discusses the threats to validity. The answers to the research questions are given in Section 8.1, and the directions that follow from the limitations are in Section 7.

6.1 Interpretation of Results

Table 6.1. Claim, supporting evidence and bound for each challenge. The last column states what the evidence does not cover.

Challenge	**Claim the evidence supports**	**Evidence (class)**	**Bound; not shown**
C1: bounded stop time	The firmware removes drive within 2042 ms of the last command, whatever the state of the operator link	Table 5.2b: 30 of 30 trials, mean 2022.5 ms (MEASURED from `p1_events.log`); F7 and one unplanned stall (Tables 5.9a, 5.9c); native test of the compiled firmware (Table 5.5d, SOFTWARE-TEST)	Time at which drive is removed, not the time the wheels stop (E4 pending); no stop distance (E3 [VERIFICATION REQUIRED]); resolution about 15 ms
C2: no single process failure ends the mission	The watchdog restored service with a new PID after SIGKILL of P1, P2 and P3 in 30 of 30 trials; the dead-man holds the motors stopped meanwhile	Tables 5.8, 5.8a (MEASURED); P1 recovery 16.03 s inside the 14.6 to 19.6 s model window (PREDICTED)	Pooled failure probability below 11.4% at 95% confidence on the bench; SIGKILL only; Arduino removal needs a P1 restart (F6)
C3: two-way victim audio and video within edge limits	An operator media session costs about 63% of one core on P2 while P1 stays near 4%; no throttling on an adequate supply	Tables 5.10, 5.10b (MEASURED)	Processor cost only; no delay, frame-rate, quality or end-to-end result (E5, E13 pending)

Challenge	Claim the evidence supports	Evidence (class)	Bound; not shown
C4: affordability	The itemised bill of materials is commodity parts totalling about USD 310	Table 5.19 (receipt-backed; BDT at an indicative rate)	Webcam, speaker and Pi power bank not in the receipts; no cost ratio computable

C1: the stop is bounded on the silence, not on the link. The strongest result of the paper is that the interval from the last command written to the Arduino to the firmware's PANIC line never exceeded 2042 ms in 30 trials, below the analytical bound of about 2060 ms (Section 3.5.2). The decision runs in the firmware's own safety task and not in a Linux process, which is the property that distinguishes this design from a stop issued by the Pi; the unit tests of Section 5.1a cannot establish it, since they exercise a model of the firmware, whereas Table 5.2b is timed against the firmware itself. The injected Wi-Fi drop shows the practical consequence: the firmware stopped 2.02 s into the outage, about 21 s before the control socket closed (Table 5.9a), so liveness was enforced by the firmware and not by the transport. The result has two limits that the measurements make plain. First, "stopped" means that drive was removed electrically; the wheels' run-down is the subject of E4 and the operator-triggered stop distance of E3 (Table 5.4), whose recorded distances are contradicted by the robot's own logs and by the 120 ms ramp bound, and therefore support no claim. Second, the guarantee holds on the silence between valid commands, so its margin against a lossy link depends on how often the dashboard sends them. At the configured 500 ms heartbeat two consecutive losses are tolerated and a third trips the dead-man (Table 5.5b, CALCULATED), but the logged interval was 1.00 s in the windows before F7 and before the unplanned stalls (Table 5.9d), at which a single loss trips it. The dead-man therefore remained safe in every observation, while its availability under loss was lower than the design value in the sessions where the dashboard throttled its heartbeat.

C2: supervision converts a loss of service into a bounded outage. All 30 SIGKILL trials recovered with a confirmed new PID, with mean recovery between 12.8 s (P3) and 16.0 s (P1), and the P1 figure falls inside the window predicted by the recovery model of Section 3.7.2. The recovery times are seconds, not milliseconds: the dominant term is the fixed cooldown of the respawn path, so the contribution of the design is not fast recovery in the sense of recovery-oriented computing [**?**] but that the outage cannot be unsafe, because the dead-man, which does not depend on any Pi process, holds the motors stopped while the process is down. The unplugged-Arduino fault (F6) marks the edge of this claim: P1 stayed up and kept serving clients, which is the isolation intended, but it did not reopen the re-enumerated device, so restoring drive after a cable fault needs a restart of P1. Process supervision as built covers crashes, not hardware faults, and Section 7.2.1 takes this up.

C3: the victim path is affordable in processor terms and isolated from control. The media process used 62.92% of one core (95% CI 62.22 to 63.67) with an operator stream against 1.12%

without, while P1 used 3.81% against 3.74% (overlapping intervals), so the cost of media lands on a process that carries no drive or stop command, which is the isolation the three-process split is designed to give. On the original power bank the same load drove the supply into under-voltage on 276 of 299 samples (Table 5.10b, MEASURED), so the "affordable" figure holds only with an adequate supply. What this evidence cannot support is any statement about the quality of the victim interaction: delay, frame rate, intelligibility and the victim's experience were not measured, and no run in Section 5 exercises the path end to end. The contribution is therefore a feasibility and cost result for C3, in line with the gap stated in Section 2.6 as a combination (two-way audio and video on single-board hardware, with its resource cost) and not as a first: two-way voice is a feature of fielded mine-rescue robots [**?**].

C4: low cost is supported by receipts, with gaps. The receipt total of about USD 310 (37,158 BDT) is dominated by the Pi with its display and storage and by the motor drive, with no specialised part. It is incomplete in a direction that matters: the USB webcam, the speaker and the Pi's power bank are absent from the record, and the replacement supply that the resource measurements made necessary is likewise not priced, so the recorded total slightly understates the full build. None of the comparison platforms reports a numeric build cost (Table 5.20), so a cost ratio cannot be computed and none is claimed.

Relation to the literature. The comparison with the platforms of Table 2.1 is qualitative and rests on what each paper reports; "NR" means not reported, not absent, and the rows differ in kind of evidence, so none of this is a controlled head-to-head. Within that limit, the pattern is consistent. TXRob, a Raspberry Pi 4 platform, sends video and microphone audio one way to the operator [**?**]; a planned speaker and microphone is not yet implemented [**?**]; and a LoRa-linked robot offers a push-to-talk walkie-talkie with no video stream [**?**]. For C1 and C2 no recent platform in the table reports a stop time bounded independently of its software or supervised recovery of its processes, whereas link-loss failsafes are documented for general-purpose autopilot software [**?**]. The measured contribution here is not that these mechanisms are new, which Section 2.3 does not claim, but that their timing and recovery have been measured together on one low-cost robot. The human-factors literature identifies command and video latency as a main degrader of teleoperation [**?**]; because no Wi-Fi latency figure exists for this robot (E2), the paper cannot yet place its control path against that literature.

6.2 Design Trade-offs and Lessons Learned

Table 6.2. Design decisions, what each bought and what each cost, with the evidence for both sides.

Decision	Gain (evidence)	Cost (evidence)
2000 ms dead-man window with a 500 ms heartbeat	Stop bounded at 2042 ms regardless of transport (Table 5.2b); two lost heartbeats tolerated (Table 5.5b)	A silent gap of 2.080 s tripped the dead-man (Table 5.9c); at a 1.00 s heartbeat a single loss trips it; SIMULATED trip rate 6.84% per 300 s mission at 5% independent heartbeat loss (Table 5.5c)
Control, media and supervision in separate processes	P1 at about 4% of one core with or without media (Table 5.10b); a media fault is designed not to reach the control path (not tested; ablation 3)	Media session costs about 63% of one core; recovery of a crashed process takes 12.8 to 16.0 s (Table 5.8); the benefit over a single process is not measured (ablation 3 not run)
One writer to the serial link, no automatic reopen of a removed device	No competing writers on the Arduino link (Section 3.6.1)	After the Arduino was unplugged and re-bound the node changed and recovery needed a P1 restart (F6)
10 m distance filter on the stored GPS track	Intended to suppress drift around a stopped robot (Section 3.10)	130 points and 1609 m of path from a robot that was not driven (Table 5.13b); the filter admits drift at 3 to 5 satellites

The table shows that each decision is a trade, not an improvement on every axis, and the paper's claims are limited accordingly. The remaining lessons are methodological and came from the build and the audit of Section 5.

A test count is not evidence about the firmware. The 146 unit tests of Section 5.1a passed while the model they exercised differed from the compiled firmware on two properties: the dead-man trips at, not after, exactly 2000 ms, and pan, tilt and status commands do re-arm it (Table 5.5d). The discrepancy was found only by compiling the unmodified firmware sources against a host-side stub. The lesson is to test the artefact that is flashed, or a build of it, and to report model-based tests under their own evidence class.

Budget the resources on the intended supply. The processor budget of the media path was first captured on a supply that could not sustain it. The throttled capture is the only one in which P2 exceeded one core (118.1%), and it was valid as a finding about the original build and invalid as a resource figure. A measurement of load must record the throttle word with it, as `reslog.py` does.

Hand-entered data needs an independent log. Rows typed after a session carry the time of entry, not of measurement, and three of the four hand-entered sheets (E3, E10, E11) showed readings that equal their references or distances no log supports. Only the robot's own logs (`p1_events.log`, `telemetry.log`) could separate sound transcription from unsupported

entry (Table 5.0), and they rotate. The practice that follows is to automate each measurement that the robot can log, to copy the log windows a claim relies on into the repository, and to treat the cross-check as part of the protocol and not as an afterthought.

Experiment configuration belongs in version control. The 16 kbit/s Opus setting existed only as an uncommitted change identified by a label, and a P2 restart between the two captures confounded the comparison (Section 5.5). A bitrate ablation therefore did not yield a result beyond the CPU figure, which showed no detectable reduction.

Instrument resolution should be chosen before the run. The telemetry log wrote coordinates to four decimal places (about 11 x 10 m), so the E9 statistics are bounded by the logging format, and the latency figures are bounded by a 500 ms telemetry cycle. In both cases a higher-resolution record existed on the robot (six decimal places in P1's memory, the P1 event log with millisecond stamps), and using it changed the conclusion.

6.3 Limitations

The limitations that determine what a reader may cite are the following. Section 4.4 lists every deviation from the plan in detail and Table 5.0 gives the evidence status of each experiment; they are not repeated here.

1. **No physical timing of the stop.** The wheel-stop time after a dead-man trip (E4) and the operator-triggered stop distance (E3) are not established. The paper claims a bounded time at which drive is removed, not a bounded stopping distance.

2. **No end-to-end performance.** True control latency over Wi-Fi (E2), media delay and quality (E5), the field trial (E13) and the usability study (E14) have no data. The paper makes no claim about operator performance, victim experience or rescue effectiveness, and no human-subject data exist; ethics approval is required before E13 or E14 is run.

3. **No measured benefit over a baseline.** The ablations of Section 4.3 that would show what the dead-man, the watchdog and the process split each contribute were not run. The paper reports the behaviour of the system with the features present and cites no without-feature measurement.

4. **Narrow fault and environment coverage.** Faults were SIGKILL crashes plus single manual injections; hangs, camera and microphone loss were not injected. The range sweep is partial and line-of-sight, and no failure point was reached. All runs were indoors or in an open area on one access point.

5. **Weak or unverified sensor evidence.** The ultrasonic accuracy (E10) and the E11 error columns are [VERIFICATION REQUIRED] and support no claim; the DHT11 was checked at one set-point and the gas sensor reports uncalibrated counts.

6. **Software-only checks are not hardware results.** The unit tests, the native firmware test and the heartbeat-loss analysis (SIMULATED) are not measurements on the assembled robot.

7. **Cost is an estimate.** No price is tied to a receipt, and the build as evaluated is not the build whose cost is stated.

What the data do support, within the bounds of Table 6.1, is the claim of Section 5.10: a firmware-enforced stop time, measured process recovery, and a measured processor cost for one operator media session, all on one robot.

6.4 Threats to Validity

Table 6.3. Threats to validity by type, the experiments they affect and the mitigation applied. Itemised deviations are in Section 4.4.

Type	**Threat**	**Affects**	**Mitigation applied; residual**
Internal	Uncontrolled captures on different days, with different uptimes and remote-desktop states	E8 comparisons	Described as descriptive, no causal effect inferred; batch-means intervals (Table 5.10b)
Internal	The two range sessions differ by about 17 dB at the same distances; the cause is not recorded	E6	Both sessions reported; no range bound claimed; cause unresolved
Internal	Driving conditions stated by the operator to have wheels clear of the ground; the logs cannot confirm it	E8	Stated as unverified; motor-current effect on the supply may be absent
Internal	Hand-entered rows carry entry time; two sheets contradicted by logs	E3, E10, E11, E12	Cross-check against machine logs (Table 5.0); log excerpts preserved; unsupported rows retained but not used
Construct	Telemetry-confirmed latencies are upper bounds that exclude Wi-Fi and can exceed the true value by up to 700 ms	E1	Labelled as upper bounds; E2 pending; log-timed dead-man interval is not subject to this quantisation
Construct	“Stop” is measured as removal of drive, not as the wheels at rest	C1	Stated wherever the figure is used; E4 pending
Construct	ICMP loss to the router is not heartbeat loss	E6, Section 5.1b	Heartbeat-loss model labelled SIMULATED; loss assumed independent, which understates trips under bursty loss

Type	Threat	Affects	Mitigation applied; residual
Construct	GPS coordinates logged to four decimals	E9	Precision stated as resolution-limited; no accuracy claim (no surveyed reference)
External	One robot, one operator, one site per experiment, one access point	All	No claim about between-unit or between-operator variation; range and loss figures stated as properties of this access point
External	Indoor, short runs (3 to 30 min), SoC at or below 52.6 deg C	E8	No claim about full missions, heat or high ambient temperature
Conclusion	Small samples (10 to 30 trials; 5 for the sensors)	E1, E7, E11, E12	95% intervals, Wilson intervals for proportions; no significance test; p95 of 30 values reported as descriptive only

Two threats deserve a comment beyond the table. First, the latency probe randomises its send phase by 0 to 200 ms so that trials are not locked to the firmware's 500 ms telemetry cycle. The phase-locked control run (Table 5.2) shows the artefact the jitter removes, medians collapsing onto the 200 ms rebroadcast grid with a deceptively small SD, so the jittered figures are the primary result; but a 200 ms jitter cannot fully decorrelate a trial from a 500 ms cycle, and the jittered spread still understates the quantisation spread. Second, the audit that produced Table 5.0 is itself a mitigation with a boundary: it checks that an entry agrees with the robot's logs, not that the entry is the result of the physical experiment described. Where the logs cannot decide (for example the wheels-clear premise), the paper states the limit instead of resolving it by assumption. Claims withdrawn during the audit are listed in the relevant subsections of Section 5 and the earlier text is preserved in `research paper/archive/`.